

Polyatomic Molecules

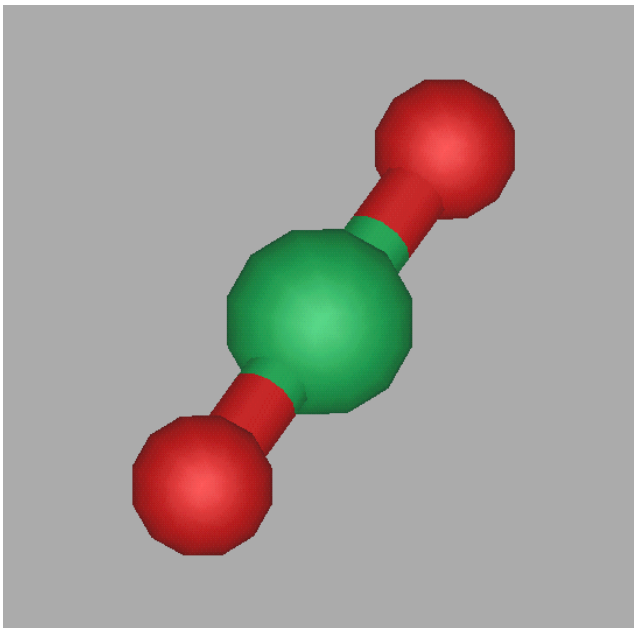
John W Daily

Polyatomic Molecules

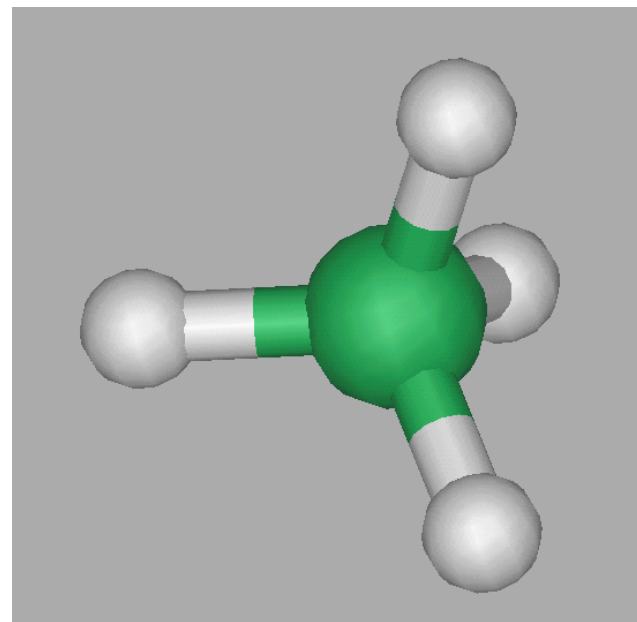
- For a body made up of n atoms, there are $3n$ degrees of freedom.
- Translation take up 3 degrees of freedom
- Rotation takes up 2 for a linear molecule, and 3 for a non-linear molecule
- Vibration then takes $3n-5$ or $3n-6$.

Rotational Modes

Rotation



CO₂ is linear
Rotates about two axis

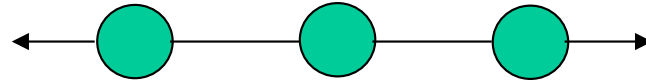


CH₄ is non-linear
Rotates about three axis

Vibrational Modes

Linear Triatomic (CO_2) - Four Modes

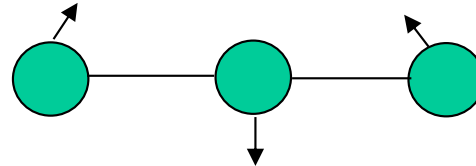
Symmetric stretch



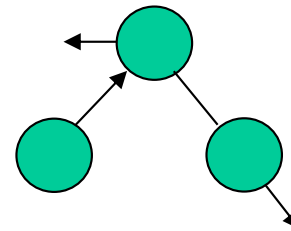
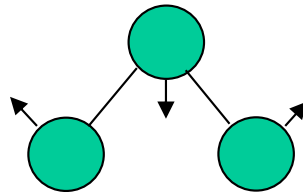
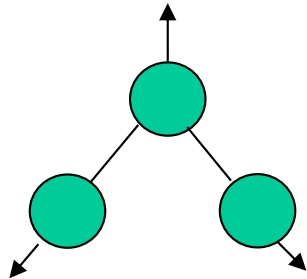
Asymmetric stretch



Bending in two planes



Non-linear Triatomic (H_2O) - Three Modes



Electronic Structure

- Atomic structure determined by quantum mechanics
- Electrons form orbitals. These combine when making a molecule into structures that are somewhat related to atomic orbitals

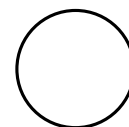
C $(1s)^2 (2s)^2 (2p)^2$

N $(1s)^2 (2s)^2 (2p)^3$

O $(1s)^2 (2s)^2 (2p)^4$

F $(1s)^2 (2s)^2 (2p)^5$

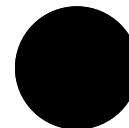
Ne $(1s)^2 (2s)^2 (2p)^6$



2s orbital (isotropic)



2p orbital (1 node)



Types of Bonds

- Ionic - Ions held together weakly by electrostatic force.
- Covalent - Electron orbits overlap. This is what holds hydrocarbons together.
 - Single - Two electrons, one from each atom
 - Double - Four electrons, two from each atom
 - Triple - Six electrons, three from each atom

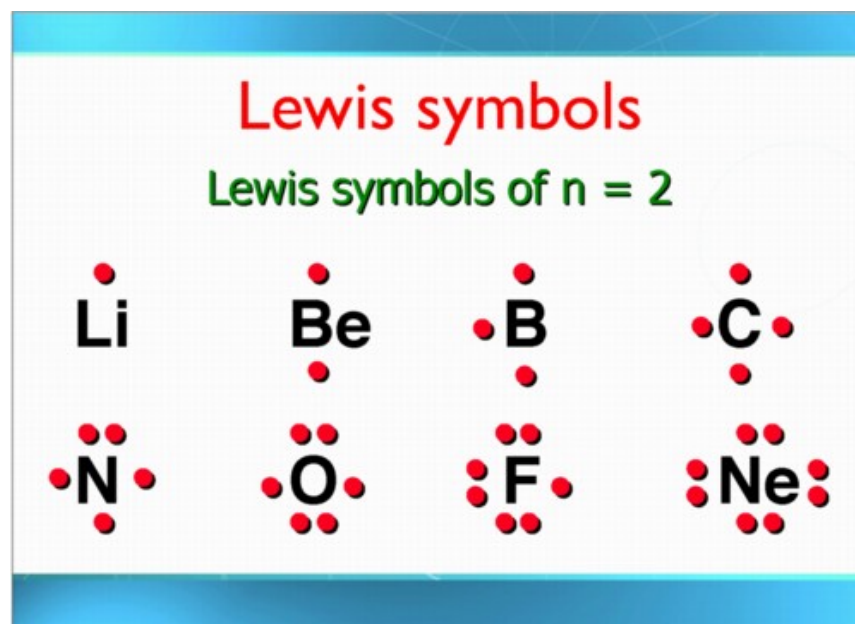
Classification of Compounds

- Hydrides
- Oxides
- Halides
- Organic vs. Inorganic
- Ionic
- Organometallic

<http://www.britannica.com/EBchecked/topic/108614/chemical-compound>

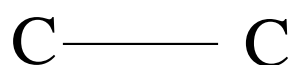
The Octet Rule

- The **octet rule** says that atoms tend to gain, lose or share electrons so as to have eight electrons in their outer electron shell
- This determines how an atom is likely to react

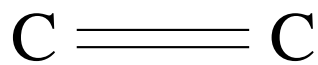


Types of Organic Compounds

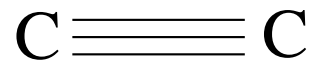
- Alkanes - Single bonds
- Alkenes - Double bonds involved
- Alkynes - Triple bonds involved
- Arines (Aromatic compounds) - Rings



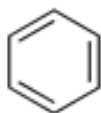
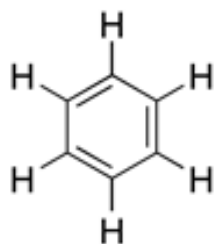
Ethane



Ethene



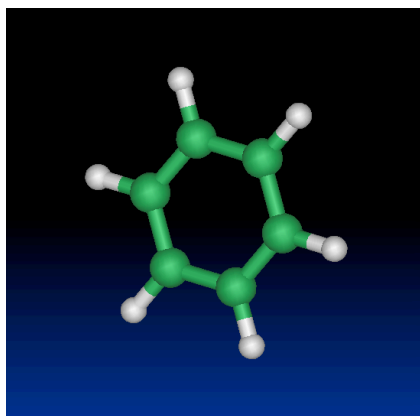
Ethyne
(Acetylene)



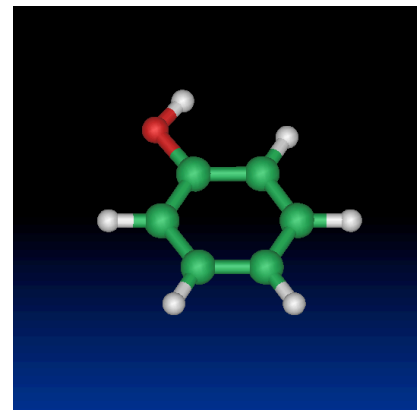
Benzene

Other Examples

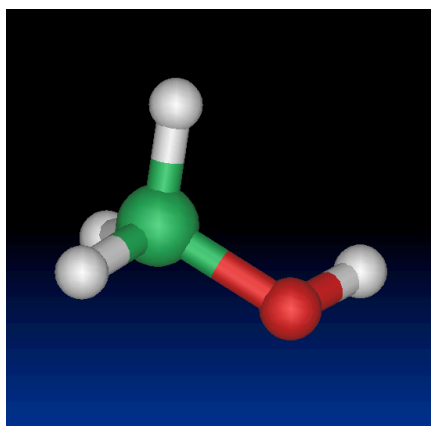
(Pick anything from NIST database)



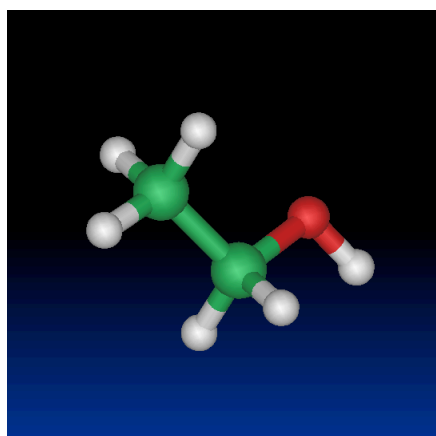
Benzene



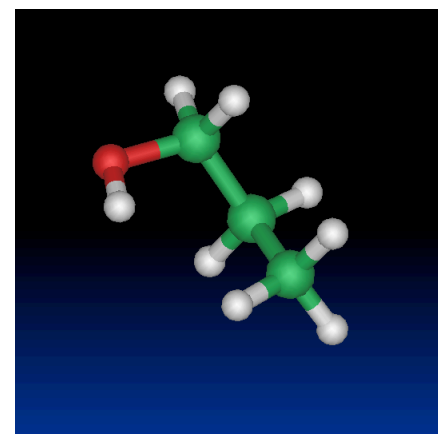
Phenol



Methanol



Ethanol



Propanol

Functional Groups

- Functional groups can replace hydrogens.
- An unlimited range of compounds can be formed.

Some Common Functional Groups	
Alkane	$-(CH_2)-$
Alkene	$\begin{array}{c} R & & R \\ & \diagdown & / \\ & C = C \\ & / & \diagdown \\ R & & R \end{array}$
Alkyne	$R-C \equiv C-R'$
Alcohol	$R-OH$
Ketone	$\begin{array}{c} O \\ \\ R-C-R' \end{array}$
Aldehyde	$\begin{array}{c} O \\ \\ R-C-H \end{array}$
Carboxylic Acid	$\begin{array}{c} O \\ \\ R-C-OH \end{array}$
Ester	$\begin{array}{c} O \\ \\ R-C-OR' \end{array}$
Amide	$\begin{array}{c} O \\ \\ R-C-NR'R'' \end{array}$
Nitrile	$R-C \equiv N$

Example - Cellulose

